Shivam Madan

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EDUCATION

BITS Pilani, Hyderabad Campus — **Hyderabad, TS**

B.E. Computer Science | CGPA: 8.32 — Expected May 2028

Relevant Coursework: Data Structures and Algorithms, Database Management Systems, Object Oriented Programming, Discrete Structures in Computer Science, Data Mining, Logic in Computer Science, Digital Design, Microprocessors and Interfacing

TECHNICAL SKILLS

Languages: C, C++, Java, Python, JavaScript/TypeScript, SQL

Frameworks: FastAPI, React, Next.js, Node.js, LangChain

Databases & Infra: MySQL, Oracle RDBMS, PostgreSQL, ChromaDB, GitHub Actions

AI/NLP: Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), Vector Embeddings, Multi-Agent Systems, Prompt Engineering

Tools & Platforms: GitHub, VS Code, Linux

WORK EXPERIENCE

Stori AI *(AI-powered Masters Admissions Platform)* — May 2025 – Present

Data & AI Engineering Intern

- Architected a multi-phase distributed data pipeline processing 200+ university web pages end-to-end across crawl, extraction, safe-merge, and live database write stages, powering a live production system used for real-time school intelligence at scale.
- Improved pipeline throughput by an estimated 60% by implementing a smarter fetch strategy that detects page types before browser instantiation, and concurrent domain fetching, validated through custom benchmarking scripts measuring per-page latency before and after.
- Built two end-to-end data acquisition pipelines from scratch, collecting and structuring 500+ real admission outcome records across multiple target programmes to bootstrap a data-driven recommendation engine.
- Improved LLM extraction accuracy across 140+ university programmes by enforcing structured JSON output and engineering bucket-specific extraction prompts, ensuring schema-compliant structured output at every stage of the pipeline.

PROJECTS

InteLect – AI-Powered Active Learning Platform — May 2025 – June 2025

- Engineered a production RAG pipeline over timestamped YouTube transcripts using ChromaDB and LangChain, enabling timestamp-grounded Q&A with 94% retrieval accuracy across 100+ transcript chunks, with optimized chunking to minimize mid-sentence cutoffs and improve retrieval precision.
- Optimized comprehension-based quiz coverage by implementing dynamic bucket-based sampling across ChromaDB chunks, guaranteeing 100% temporal coverage across all video segments.
- Deployed a distributed full-stack system with a FastAPI backend (Railway) and Next.js + TypeScript frontend (Vercel), handling concurrent API requests across transcription, vector retrieval, and LLM inference endpoints.

Sieve – AI Recruitment Intelligence Platform — June 2025 – Present

- Built a 9-agent crewAI pipeline that automates end-to-end internship screening across JD analysis, resume scoring, deep candidate profiling, interview question generation, and personalised email drafting via async FastAPI backend
- Reduced agent hallucination rate by 73% by identifying and resolving inconsistent scoring scales across LLM outputs, enforcing uniform per-field constraints via prompt engineering to achieve standardized, normalized scores across all candidates.
- Engineered a recruiter calibration system that learns individual scoring preferences from manual evaluations, dynamically refining prompts through a validation loop benchmarked against human recruiter judgments to continuously align automated scoring with recruiter preferences.

EXTRA CURRICULAR

Core Member, Department of Professional Events (DoPE)

- Coordinated the planning and execution of large-scale festival pro-shows attended by 3,000+ students, collaborating with cross-functional teams and artist representatives to manage logistics, operational requirements, and on-ground event delivery.

Core Member, Communications and Guest Relations Team, Embryo

- Managed outreach and coordination for 10+ prominent speakers across entrepreneurship, academia, public service, and entertainment, facilitating sessions attended by 500+ participants.